Exposure-Adjusted Pedestrian Harm Is 16 Percent Higher per 10 Percentage Points of Neighborhood Poverty in New York City

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Abstract

Pedestrian injury and fatality rates are frequently reported to be elevated in lower-income neighborhoods, but population-based rates confound infrastructure hazards with variations in pedestrian exposure. We integrate modeled pedestrian volumes for New York City with street centerline geometry and five years of geocoded police-reported collision records (2019–2023). Crosswalk segments are identified and clustered into 41,937 discrete crossing sites, predominantly intersections, to which collisions are spatially assigned. Across 38,477 sites located in census tracts with complete covariate data, pedestrian injury and fatality collisions increase by 16% per 10-percentage-point increase in the tract poverty rate (rate ratio [RR] 1.16, 95% CI [1.12, 1.20]), conditioning on pedestrian crossing volume, roadway width, lane counts, posted speed limits, tract-level commuting mode shares, land use, and borough fixed effects. A model that holds only pedestrian volume fixed yields an RR of 1.30 (95% CI [1.26, 1.35]). This gradient is robust to alternative count estimators, site grouping distances, collision assignment radii, temporal partitions, borough exclusions, clustered covariance structures, and a designated truck route control. The estimate attenuates to 1.10 with ZIP-code fixed effects and increases to 1.19–1.22 after correcting for American Community Survey sampling error in poverty estimates. Pedestrian fatalities show a similar, somewhat steeper gradient (RR 1.29, 95% CI [1.17, 1.42]), arising from both higher collision frequency and a higher chance that a collision is fatal, although the second component is imprecise. Conventional models relying on residential population yield attenuated ratios of 1.06–1.09, indicating that the observed disparity is uncovered by exposure adjustment rather than by spatial aggregation. These estimates reflect observational associations in the absence of vehicle volume counts. An open-source pipeline is provided to reproduce all data processing, models, and figures from public records.

Keywords: pedestrian safety; traffic crashes; neighborhood poverty; pedestrian exposure; New York City; Poisson regression

1 Introduction

Between 2019 and 2023, the New York City Police Department recorded 37,815 crashes away from highways in which a pedestrian was injured or killed, including 469 pedestrian deaths. These harms are distributed unevenly throughout the city, and a common method to display the unevenness is to divide crash counts by the number of residents in each neighborhood.

Residential population is an incomplete exposure measure, however. Pedestrians travel along commercial corridors, between transit stops, and to places of work, not only within the areas where they live. A lower-income neighborhood can therefore show a high crash rate per resident because its streets are more hazardous, because more people cross them, or for both reasons (and only the first reason is a difference in safety). Separating these explanations requires a measure of how many people walk across each crossing.

Recent network-wide estimates of pedestrian volume for New York City [17] make such a measure available. We use these modeled volumes to quantify exposure at 38,477 crossing sites, most of them intersections, and we ask what neighborhood poverty adds once local exposure is held fixed.

Specifically, we ask whether neighborhood poverty remains associated with pedestrian harm after accounting for pedestrian volume and street design. We report two disparities. The total gap is the association that remains with only crossing volume and street design (width, lanes, and posted speed) held fixed. The residual gap is the association that remains after tract commuting patterns, commercial land use, and borough are also held fixed. The two gaps answer different questions, because transit use and walking may lie on the causal path from poverty to exposure to harm.

This paper makes three contributions.

1. We measure exposure at 38,477 crossing sites rather than per resident, and we show that exposure-adjusted pedestrian harm is about 16% higher per 10 percentage points of tract poverty (30,866 harm crashes and 411 deaths).
2. We test the stability of this estimate against the choice of estimator, the definition of a site, spatial

dependence, a truck route proxy for vehicle traffic, sampling error in the poverty rate, error in the modeled volumes, and the choice of geographic scale.

3. We release a tested, open-source package that rebuilds every number from public sources and two small processed inputs.

2 Related Work

Exposure in pedestrian safety models. Pedestrian counts are scarce, so many safety performance functions omit pedestrian exposure or replace it with proxies such as population density or the share of commuters who walk [18]. Tao et al. compared Minneapolis models with and without estimated pedestrian and bicycle exposure, entered in logarithmic form alongside vehicle traffic. Including exposure improved model fit and changed which locations ranked as highest risk. For pedestrians, only 42% of the top 1% of intersections were common to both sets of models. For New York City, the NYC Walks authors divided pedestrian injuries at intersections by modeled volume and mapped the result, finding that the intersections with the highest injury risk per pedestrian are often outside Manhattan [17]. That analysis is descriptive and does not relate risk to neighborhood characteristics.

Neighborhood disparities. The New York City Department of Health reported a pedestrian fatality rate of 9.1 per 100 miles of street in ZIP codes with very high poverty, against 3.2 in low-poverty ZIP codes, for 2012 to 2014, when 497 pedestrians died citywide [10]. The denominator in this is street length, which does not indicate how many people walk. The city's current equity reporting under Vision Zero likewise uses rates per resident, comparing counts of people killed or severely injured per 1,000 population in priority investment areas, which are defined largely by their share of low-income and non-White residents [6]. In Oregon, rates of fatal and severe pedestrian injury were nearly three times higher in census tracts with the highest concentration of poverty and people of color than in those with the lowest [15]. This Oregon analysis is descriptive. It compares categories and reports higher arterial traffic, more transit stops, and more walking and transit commuting in the high-concentration tracts. In Minneapolis, modeled pedestrian crash risk at intersections was higher in areas of concentrated poverty than elsewhere, with exposure included in the models [5]. That comparison relies on simple tests of means across areas and on exposure estimated from counts at a few hundred locations.

Vision Zero in New York City. The Department of Transportation's equity report compares pedestrian fatalities in 2014 through 2023 with those in 2004 through 2013 across Neighborhood Tabulation Areas grouped by poverty [12]. Pedestrian fatalities fell in every poverty group, by 34% in the highest-poverty areas and by 24% to 25% in the others, and street improvement projects were more frequent per street mile in high-poverty areas. These results describe change over time in fatality counts and do not compare the level of harm across areas, and they do not adjust for how many people walk. The department's borough action plans identify priority corridors, intersections, and areas from counts of pedestrians killed or severely injured [11], which rest on crash counts alone. Our analysis asks a different question, namely whether harm per crossing site still rises with neighborhood poverty once pedestrian volume is held fixed.

Contribution. We bring the two literatures together at a larger scale. We estimate the poverty gradient in pedestrian harm at 38,477 crossing sites in a single city, with modeled pedestrian volume at every site, and we report how the gradient changes as exposure, street design, and neighborhood context are added. We also test the estimate against sampling error, spatial dependence, a proxy for vehicle traffic, and the choice of geographic scale, which is known to affect area-based results [4, 14]. We lack vehicle volumes, which the Minneapolis models include.

3 Data

3.1 Crashes

Outcomes are sourced from the NYC Motor Vehicle Collisions Crashes file [13]. We retain crashes from 2019 through 2023 in which at least one pedestrian was injured or killed, which we call pedestrian harm crashes. We drop crashes flagged as occurring on highways and keep those with valid coordinates. This leaves 37,815 crashes in which 469 pedestrians died.

3.2 Pedestrian volumes

Exposure comes from NYC Walks [17], which models peak-hour pedestrian volumes on 315,577 sidewalk, crosswalk, and footpath segments. The model is calibrated to New York City Department of Transportation counts from 2018 and 2019. The file provides volumes for six peak hours (weekday and weekend, each in the morning, midday, and evening). We average the six to form one volume index per segment. This index measures relative exposure and it is not a daily count. Because the volumes date from 2018 and 2019 while crashes run to 2023, the index assumes that the relative pattern of walking across the city changed little, which the pandemic disruption of 2020 makes questionable. We return to this point in the limitations.

3.3 Streets, tracts, and tract covariates

Street width, travel lanes, posted speed, road type, and truck route class come from the NYC street centerline file [8]. Tract poverty, population, walking share of commuting, and transit share of commuting come from the 2019 to 2023 American Community Survey (ACS) five-year estimates [19]. The commercial share of floor area comes from PLUTO [9]. Tracts follow the 2020 cartographic boundaries. The tract table has 2,189 tracts, and 2,183 of them contain a site.

4 Methods

4.1 Inferring crosswalks and sites

The NYC Walks file has no field that identifies crosswalks, so we infer them as pedestrian segments that intersect the centerline of an ordinary street, meaning road type 1, active status, and not marked non-vehicular. This results in 110,603 segments, or 35% of the network. Their median length is 16.6 m, against 64.6 m for all other segments, which is consistent with crosswalks being short crossings and with sidewalks and paths being longer. We treat crosswalk midpoints within 25 m of each other as a single site, using DBSCAN with a minimum of one point. This produces 41,937 sites with a median of three crosswalks per site. Since crash coordinates in the source file tend to lie at the intersection center, a site (instead of a single crosswalk) is treated as the appropriate unit. Figure 1 maps the sites.

Each site takes the sum of its crosswalk volumes, the widest street it crosses, the highest posted speed, and the largest number of travel lanes. It also takes the poverty rate and other covariates of the tract that contains it.

4.2 Assigning crashes

We assign each crash to the nearest site within 30 m. This rule captures 81.6% of harm crashes (30,866 of 37,815) and 87.6% of pedestrian deaths (411 of 469). Most of the remaining crashes occur mid-block. The results therefore describe harm near crossings, and we report sensitivity to radii of 20 and 50 m.

4.3 Models

For site i in tract $t(i)$ and borough $b(i)$, let Y_i be the crash count over the five years. The primary model is a Poisson regression with a log link,

$$\log \mathbb{E}[Y_i] = \beta_0 + \beta_P \, P_{t(i)} + \gamma \log V_i + \boldsymbol{\delta}^\top \mathbf{S}_i + \boldsymbol{\theta}^\top \mathbf{Z}_{t(i)} + \eta_{b(i)}, \tag{1}$$

where P is the tract poverty rate in percentage points, V_i is site volume, $\mathbf{S}_i$ holds street width in units of 10 ft, travel lanes, and posted speed in units of 5 mph, $\mathbf{Z}$ holds crosswalks per site and the standardized walking, transit, and commercial shares, and η_b is a borough effect. The quantity of interest is $\mathrm{RR} = \exp(10\beta_P)$, the rate ratio per 10 percentage point increase in poverty. Entering log volume as a covariate allows risk to rise more or less than proportionally with exposure, which a fixed offset would rule out. The coefficient γ is the volume elasticity.

We estimate four specifications to show how the ratio changes as controls enter.

A Poverty and log volume.

B Model A plus street width, lanes, and speed.

C Model B plus crosswalks per site and tract controls.

D Model C plus borough effects. This is the primary specification.

Poisson estimates remain consistent under overdispersion [16], so we use them as the primary estimator and report standard errors clustered by ZIP Code Tabulation Area (183 clusters), which we call ZIP areas [2]. We cross-check using a negative binomial GLM with a moment-based dispersion parameter. We use the same specifications for pedestrian deaths. To split the death gradient, we model harm crashes per site (frequency) and deaths per harm crash among sites with a crash (severity) separately.

4.4 Robustness checks

We run the following checks, each with model D unless noted.

- *Definitions.* Site grouping distances of 15, 25, and 40 m, crash assignment radii of 20, 30, and 50 m, and removal of the top 1% of sites by crash count.
- *Time and place.* Separate estimates for two periods and for each year, using the same volumes, and estimates that leave out each borough in turn.
- *Volume error.* Simulation extrapolation [3] at assumed reliabilities of 0.95 to 0.70 for log volume, which represents the classical-error worst case.
- *Spatial dependence.* Moran's I [7] of residuals using 8 nearest neighbors, standard errors clustered by tract and by 1, 2, and 4 km grid cells, and a block bootstrap over 2 km blocks.
- *Unmeasured regional factors.* ZIP area fixed effects, which use only variation among tracts in the same ZIP area.
- *Vehicle traffic.* The data have no vehicle volumes. As a partial proxy, we flag sites on a designated truck route, add the flag as a covariate, and split the sample by it.
- *Sampling error in poverty.* The ACS reports a margin of error for each tract. We treat the margin divided by 1.645 as a known standard error and extrapolate the coefficient back to zero error.
- *Geographic scale.* We aggregate the same data to

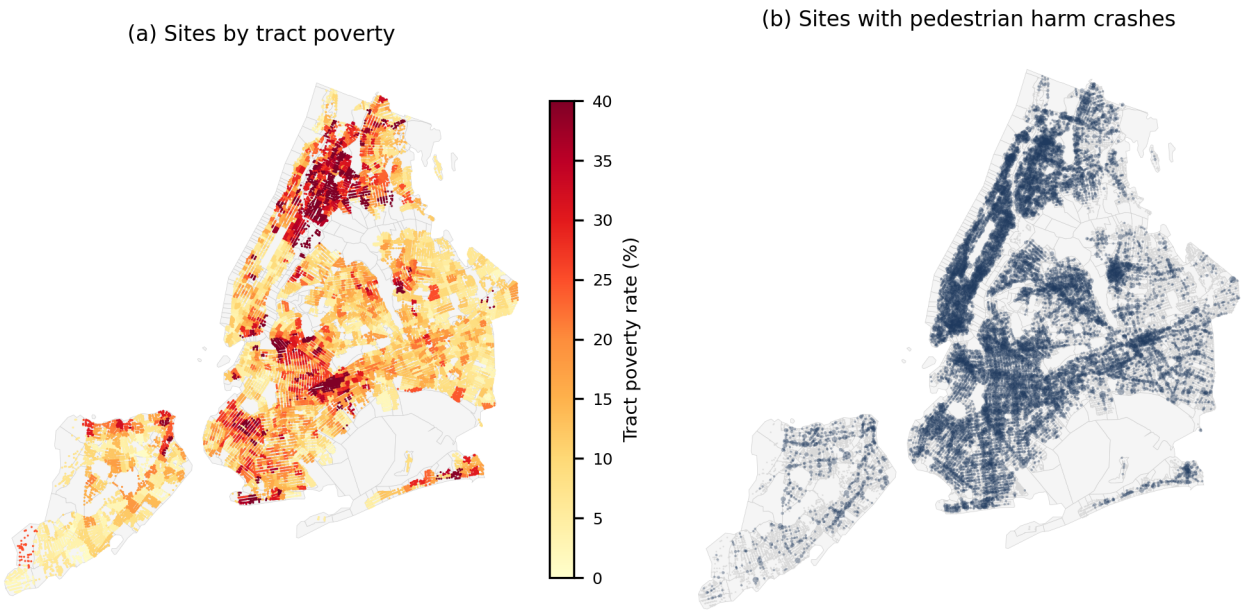


Figure 1: Study sites. (a) Each of the 38,477 model sites, colored by the poverty rate of its tract. (b) Sites with at least one pedestrian harm crash from 2019 through 2023, sized by the number of crashes.

tracts, ZIP areas, and 100 random contiguous zonings at each of five target sizes [4], and we compare two model forms. One uses residents only (log population plus walking, transit, and commercial shares and borough). The other adds volume and street design.

4.5 Use of generative AI

Generative AI was used throughout this work, in accordance with the policy of this platform. Claude, a large language model developed by Anthropic, assisted with planning the analysis, writing and testing the `nycped` software package and its unit tests, executing and checking intermediate results, and drafting and revising the manuscript text. The author specified the research question and the analytic decisions, ran the full pipeline on public data, reviewed the code, the outputs, and the manuscript, and verified every reported number against the generated outputs. The author takes full responsibility for the content of this paper, including the results and any remaining errors. No AI tool is listed as an author.

5 Results

5.1 Plausibility of the volumes

To check whether the modeled volumes track activity, we compared them with independent tract measures. Median volume per crossing is highest in Manhattan (290) and lower in Brooklyn (162), the Bronx (159), Queens (127), and Staten Island (110). Across tracts, the rank correlation of mean volume per crossing is 0.60 with population density, 0.51 with the ACS walking share, and 0.20 with poverty. The spread across boroughs seems narrow relative to density, so the volumes may be overstated in low-activity areas. We return to this under measurement error.

5.2 Pedestrian harm and poverty at crossing sites

Table 1 and Figure 2 show the poverty rate ratio as controls enter. Figure 3 shows the same pattern from models that omit poverty, as observed over expected harm by poverty decile. Sites in the poorest decile (mean tract poverty 38.5%) have 3.2 times the harm that crossing volume and street design predict, relative to sites in the least poor decile (2.7%). This ratio becomes 1.7 once tract context and borough are also fixed (95% intervals 2.65 to 3.89 and 1.44 to 1.97). With only crossing volume held fixed, harm crashes are 30% higher per 10 percentage points of poverty (1.30, 1.26 to 1.35). Street design alters the ratio only slightly (1.28). Tract commuting and land use reduce it to 1.15, and borough effects leave it at 1.16 (1.12 to 1.20). Together, the controls absorb more than 40% of the log gap, almost all of it through the tract controls. Street design alone absorbs about 8%. The negative binomial cross-check gives the same ratios within 0.06.

In model D, wider streets carry more harm. Harm

Table 1: Poverty rate ratio per 10 percentage point increase in tract poverty, with 95% confidence intervals. Poisson regression with standard errors clustered by ZIP area. Harm crashes: 38,477 sites, 30,866 crashes. Deaths: 411 deaths.

Model	Harm, Poisson	Harm, neg. binomial	Deaths, Poisson
A volume only	1.30 (1.26 to 1.35)	1.36 (1.30 to 1.41)	1.28 (1.18 to 1.39)
B + street design	1.28 (1.23 to 1.32)	1.33 (1.27 to 1.39)	1.26 (1.16 to 1.37)
C + tract controls	1.15 (1.11 to 1.18)	1.15 (1.11 to 1.19)	1.23 (1.11 to 1.36)
D + borough (primary)	1.16 (1.12 to 1.20)	1.17 (1.12 to 1.21)	1.29 (1.17 to 1.42)

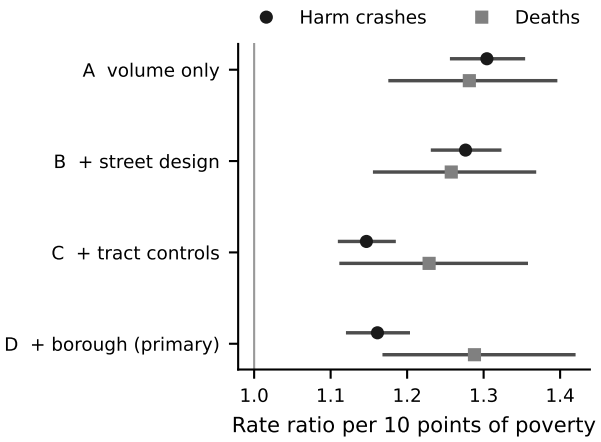


Figure 2: Poverty rate ratio per 10 percentage points as controls enter, for harm crashes and deaths. Bars are 95% intervals clustered by ZIP area.

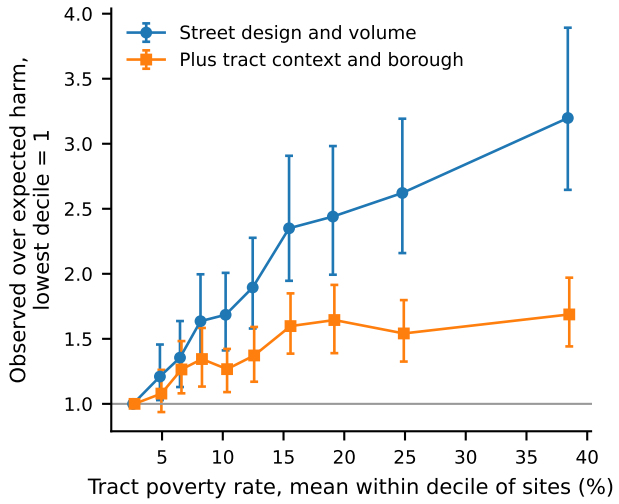


Figure 3: Observed over expected pedestrian harm crashes by decile of tract poverty, scaled so that the lowest decile equals 1 by construction and has no interval. Expected counts come from Poisson models that omit poverty. Bars are 95% intervals from resampling ZIP areas.

rises 26% per additional 10 ft of street width (1.26, 1.21 to 1.31), 9% per travel lane (1.09, 1.04 to 1.15), and 24% per 5 mph of posted speed (1.24, 1.08 to 1.41). Posted speed varies little across the city, so the speed estimate deserves caution. The volume elasticity is 0.79, meaning that harm rises less than proportionally with pedestrian volume. Standardized tract transit share has a large coefficient (1.40 per standard deviation, 1.34 to 1.46), walking share a moderate one (1.11), and commercial share a small negative one (0.93). The poverty gradient does not vary with volume (interaction $p = 0.50$) and is slightly smaller on wider streets, by about 3% per 10 ft.

5.3 Robustness

Figure 4 summarizes the checks made. The ratio stays between 1.10 and 1.20 in every check that uses the same outcome and covariates. Changing the crash radius to 20 or 50 m gives 1.17 and 1.16. Regrouping crosswalks at 15 or 40 m gives 1.14 and 1.16. Dropping the top 1% of sites gives 1.13. The two periods, 2019 to 2021 and 2022 to 2023, both give 1.16, and the yearly ratios run from 1.12 to 1.19. Leaving out each borough in turn gives 1.13 to 1.19, with the lowest value when Manhattan is excluded.

Measurement error in volumes. Suppose the modeled volumes carry classical error with reliability from 0.95 down to 0.70. The extrapolated poverty ratio then rises to 1.17 to 1.18 under linear extrapolation and to 1.18 to 1.25 under quadratic extrapolation, so noisy volumes do not inflate the primary estimate. Classical error is a worst case. Model predictions may behave more like Berkson error, which does not bias the coefficient.

Spatial dependence. Residuals are positively autocorrelated (Moran's I 0.088 at sites, permutation $p = 0.005$, and 0.198 when summed by tract), which indicates that a spatially structured factor remains unmodeled. Inference does not depend on it. Clustering by tract, or by 1, 2, or 4 km grid cells, gives intervals within 0.01 of the primary interval, and a block bootstrap over 247 blocks gives 1.16 (1.13 to 1.20).

Differences between ZIP areas. With ZIP area fixed effects, the ratio is 1.10 (1.07 to 1.13). About a

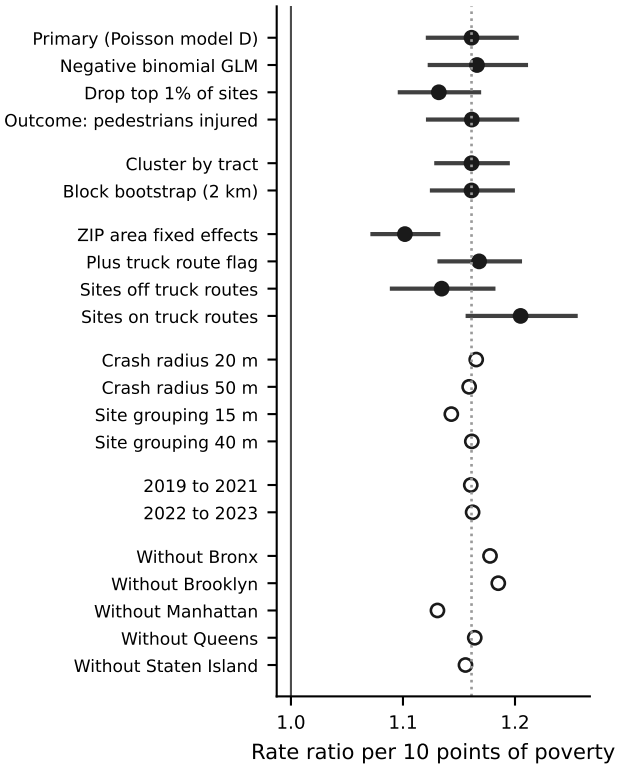


Figure 4: Poverty rate ratio per 10 percentage points across robustness checks. Bars are 95% intervals where the check reports one. Open circles are point estimates. The dotted line marks the primary estimate of 1.16.

third of the log gap therefore lies between ZIP areas, and a gradient remains among tracts within the same ZIP area. ZIP areas are large, so unmeasured neighborhood factors can still operate inside them.

Vehicle traffic. Truck route sites carry 1.47 times the harm of other sites after the other controls. Adding the flag leaves the poverty ratio at 1.17 (1.13 to 1.20). The ratio is 1.13 (1.09 to 1.18) off truck routes and 1.20 (1.16 to 1.25) on them. About 17.7% of sites qualify. The flag identifies a street class and is not a vehicle count, so this result shows only that heavy-vehicle street class does not explain the gradient.

Sampling error in poverty. The mean standard error of the tract poverty rate is 4.78 percentage points. The reliability of the poverty rate is 0.79, and 0.69 among tracts that are alike on the other controls. Correcting for this error raises the harm ratio from 1.16 to between 1.19 and 1.22 and the death ratio from 1.29 to between 1.34 and 1.41. Sampling error therefore pulls the estimate toward 1, and the primary estimate is conservative. The method assumes independent classical errors. The walking and transit shares are also ACS estimates, so some confounding that they would absorb may remain.

5.4 Pedestrian deaths

Deaths show a similar and slightly steeper gradient (Table 1). In model D, with 411 deaths, the ratio is 1.29 (1.17 to 1.42). The negative binomial GLM gives 1.28 (1.16 to 1.41), and a cluster bootstrap over 183 ZIP clusters gives 1.29 (1.16 to 1.42). Estimates range from 1.23 to 1.35 across site definitions, crash radii, periods, and borough exclusions. The ratio is 1.25 (1.13 to 1.39) for 2019 to 2021 and 1.35 (1.16 to 1.56) for 2022 to 2023, and yearly ratios run from 1.10 to 1.43 with overlapping intervals.

The frequency and severity split gives a harm crash ratio of 1.16 (1.12 to 1.20), a ratio of deaths per harm crash of 1.11 (1.00 to 1.23), and a ratio of deaths per site of 1.29 (1.17 to 1.42). About 60% of the log death gradient thus comes from more crashes and about 40% from a higher chance that a crash is fatal. The severity interval touches 1, so the second part is suggestive. About 12% of deaths (469 against 411) fall outside the 30 m site assignment, so this result describes deaths near crossings.

5.5 Geographic scale

Table 2 aggregates the same data to larger zones. With residents only, the poverty ratio for harm crashes is 1.06 to 1.09 at every scale. With volume and design, it is 1.13 at tracts, 1.20 at ZIP areas, and 1.15 to 1.18 across random zonings. For a given zone size, 95% of random zonings span about plus or minus 0.04. The exposure adjustment, and not the geography, therefore produces the larger estimate. For deaths, the exposure-adjusted ratio drifts from 1.29 at sites and 1.23 at tracts to about 1.15 for zones of 14 tracts, and the spread across zonings widens because few deaths fall in each zone.

6 Discussion

Principal finding. After accounting for how many people walk across each crossing and for the design of the street, pedestrian harm crashes are about 16% higher per 10 percentage points of tract poverty, and deaths are about 29% higher. The estimate is stable across the checks we ran and is conservative with respect to sampling error in the poverty rate. Pedestrian exposure and the geographic unit of analysis affect the result in different manners. With residents alone, the ratio is 1.06 to 1.09 at every scale, and adding crossing volume and street design raises it to 1.13 to 1.20. Total crossing volume is negatively correlated with poverty across tracts (Spearman -0.16), so poorer tracts carry less pedestrian activity and more harm. A comparison per resi-

Table 2: Poverty rate ratio per 10 percentage points at different zone scales. Poisson regression. "Residents only" uses log population, commuting shares, and borough. "With exposure" adds log crossing volume and volume-weighted street width, lanes, and speed. Random zones show the median over 100 draws.

	Harm crashes		Deaths	
Scale	Residents only	With exposure	Residents only	With exposure
Tracts (2,183)	1.06 (1.02 to 1.10)	1.13 (1.10 to 1.16)	1.16 (1.05 to 1.28)	1.23 (1.12 to 1.36)
ZIP areas (183)	1.09 (1.01 to 1.16)	1.20 (1.12 to 1.29)	1.17 (1.00 to 1.36)	1.25 (1.05 to 1.50)
Random, about 2 tracts	1.07	1.15	1.15	1.24
Random, about 3 tracts	1.06	1.16	1.12	1.20
Random, about 5 tracts	1.07	1.17	1.13	1.20
Random, about 9 tracts	1.07	1.18	1.11	1.19
Random, about 14 tracts	1.07	1.18	1.07	1.15

dent therefore understates the difference in risk at the crossing.

Two disparities. The total gap (1.28, with volume and street design held fixed) and the residual gap (1.16, after tract context and borough) relate to different inquiries. The total gap describes how much more harm a pedestrian faces at a crossing in a poorer tract than at one with the same volume and street design in a less poor tract. The residual gap asks the same question among tracts that are alike in commuting patterns and land use. Transit share has the largest coefficient in the model (1.40 per standard deviation). It may represent a mediating pathway, if residents of poorer tracts rely more on transit and so walk to and from stops along arterial streets, or it may stand in for density and the character of the street environment. The data cannot separate these readings, so we report both gaps. About a third of the log gap lies between ZIP areas and the rest lies among tracts within them, which suggests that the gradient is not only a difference between broad regions of the city.

Fatal crashes. The gradient for deaths is similar to that for harm crashes and somewhat steeper. About 60% of it reflects more crashes and about 40% a higher probability that a crash is fatal, although the second share is imprecise because its interval includes 1. The data do not allow us to explain a higher fatality rate. Possible explanations include features of the street or vehicle mix that our measures do not capture, and differences in the time taken to receive care. Each would require data that we do not have.

A modeling lesson. A per-resident tract model can miss a gradient that exposure would reveal. A tract model of deaths with population as an exact offset and no borough effects gave a poverty ratio of 1.02 (0.92 to 1.12). With population entering freely instead, tract-level death ratios are 1.07 to 1.11 in the appendix models (Appendix A) and 1.16 in the scale analysis (Table 2), and the ratio is 1.23 once crossing volume and street design are added. The earlier null was fragile to specification, and we do not present it as a finding. An offset assumes that risk is proportional to its denominator. The volume elasticity of 0.79 suggests that this assumption does not hold for pedestrian volume. However, that estimate is itself sensitive to measurement error. When we correct for classical error in log volume at assumed reliabilities of 0.95 to 0.70, the corrected elasticity runs from 1.0 to 1.6 under quadratic extrapolation, so the data do not support a conclusion that risk per pedestrian falls as volume increases. The poverty ratio is much less sensitive to the same correction (1.17 to 1.25).

Limitations.

- These are associations. The analysis has no vehicle traffic volumes and uses a truck route flag as a partial proxy. Residual confounding is possible, and residual spatial autocorrelation indicates that a spatial factor remains unmodeled.
- The pedestrian volumes are modeled, cover six peak hours, and date from 2018 and 2019, while crashes run to 2023. They were calibrated mostly at commercial corridors. Pedestrian activity changed after 2020, although the poverty ratio is stable across periods.
- About 18% of harm crashes and 12% of deaths fall outside the 30 m site assignment, mostly mid-block, and we exclude them.
- Tract measures are ecological and say nothing about who exactly is injured.
- Poverty comes from one ACS period and is measured with error. The transit and walking shares may be mediators, so the residual gap should not be read as the total effect of poverty.
- We infer crosswalks from geometry and do not observe them.

7 Conclusion

Pedestrian harm in New York City is higher at crossings in poorer neighborhoods, and the difference persists after accounting for modeled pedestrian volume, street width, travel lanes, posted speed, neighborhood commuting and land use, and borough. At the crossing level, the ratio is 1.16 per 10 percentage points of tract poverty for harm crashes and 1.29 for deaths, and it is larger once measurement error in the poverty rate is taken into account. Per-resident analyses understate this finding. Our results are associations. They do not identify a cause or the effect of any intervention, and they rest on modeled volumes from 2018 and 2019 and on a proxy for vehicle traffic. Future work should add vehicle volumes and current pedestrian counts, test whether the pattern holds in other cities, and examine which features of the street environment account for the residual gap. The package released with this paper allows each of these steps to be run on the same footing.

Code and data availability

All analysis code is in the `nycped` Python package [1], with unit tests, a command line, and a five step reproduction guide. The code is archived on Zenodo (`https://doi.org/10.5281/zenodo.23754868`). The package rebuilds the site table from three public sources (the NYC Walks supplementary network, the NYC street centerlines, and Census tract boundaries) and two small processed inputs, a tract table and a table of pedestrian harm crashes. The repository includes the script that regenerates both inputs from public sources. The NYC Walks files are not redistributed, since no license is stated. Readers should download them from the source cited in Sevtsuk et al. [17].

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Conflicts of Interest

The author declares no conflicts of interest.

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A Tract-level models

Table 3 gives the tract-level models, with crashes counted in tracts. Rate ratios are per 10 percentage points of poverty, from negative

binomial regression with standard errors clustered by ZIP area. Models 1 and 2 use residents as the exposure offset, and models 3 and 4 use crossing volume. Models 5 to 7 enter both as covariates. Controls are the tract commuting and land use measures. The data reject proportional risk, so we do not rely on them.

Table 3: Tract-level poverty rate ratios per 10 percentage points for pedestrian injuries and deaths, with 95% intervals.

Model		Injuries	Deaths
1	Residents offset, controls	1.05 (1.02, 1.08)	1.02 (0.92, 1.12)
2	Residents offset, none	1.12 (1.08, 1.17)	1.00 (0.92, 1.09)
3	Volume offset, controls	1.21 (1.16, 1.25)	1.17 (1.06, 1.28)
4	Volume offset, none	1.34 (1.28, 1.39)	1.19 (1.10, 1.30)
5	Both as covariates, controls	1.11 (1.08, 1.15)	1.08 (0.99, 1.18)
6	Model 5 plus borough	1.11 (1.08, 1.15)	1.11 (1.01, 1.22)
7	Model 5 without Staten Island	1.12 (1.08, 1.15)	1.07 (0.98, 1.18)

In models 5 and 6 the elasticities for injuries are 0.46 for population and 0.41 for crossing volume.